

YOLO Dataset & Configuration Files

Step 1: Setting Up the Environment

A virtual environment was created to isolate dependencies and ensure a clean workspace for running YOLO models.

```
venv\Scripts\activate  
pip install ultralytics
```

Step 2: Understanding YOLO Dataset Structure

The dataset used in YOLO follows a well-defined folder structure. This structure ensures proper mapping between images and their corresponding labels.

```
dataset/  
├── images/  
│   ├── train/  
│   └── val/  
├── labels/  
│   ├── train/  
│   └── val/  
└── data.yaml
```

Each image in the images folder has a corresponding label file in the labels folder.

Step 3: Exploring the YAML Configuration File

The data.yaml file acts as the central configuration file for the dataset.

Example:

```
train: images/train  
val: images/val  
nc: 3  
names: ["person", "car", "dog"]
```

Key Components:

- train → Path to training images
- val → Path to validation images
- nc → Number of classes
- names → List of class labels

This file connects the dataset to the YOLO model during training and inference.

Step 4: Understanding Label Files

Each image has a corresponding .txt file containing object annotations.

Detection Format:

class_id x_center y_center width height

Example:

0 0.52 0.48 0.30 0.40

Explanation:

- class_id → Object category index
- x_center, y_center → Center of bounding box (normalized)
- width, height → Size of bounding box (normalized)

Step 5: Segmentation Annotation Format

For segmentation tasks, labels include multiple points representing object boundaries.

class_id x1 y1 x2 y2 x3 y3 ...

This provides pixel-level information instead of just bounding boxes.

Step 6: Observations on Configuration Files

From analyzing the dataset and configuration files, the following insights were observed:

- The YAML file defines dataset paths and class mappings
- Label files store object positions in normalized format
- Proper folder structure is essential for model training
- Segmentation provides more detailed information compared to detection

Step 7: Conclusion

This task helped in understanding how YOLO organizes datasets using configuration and annotation files. The YAML file acts as a bridge between the dataset and the model, while label files provide detailed object information. This structured approach ensures efficient training and accurate predictions.

References

- Ultralytics YOLO Documentation: <https://docs.ultralytics.com>

- YOLO Dataset Format: <https://docs.ultralytics.com/datasets>
- Image Segmentation Overview: <https://www.ultralytics.com/glossary>